

Playwright Stabilization Assessment Report

Submitted By

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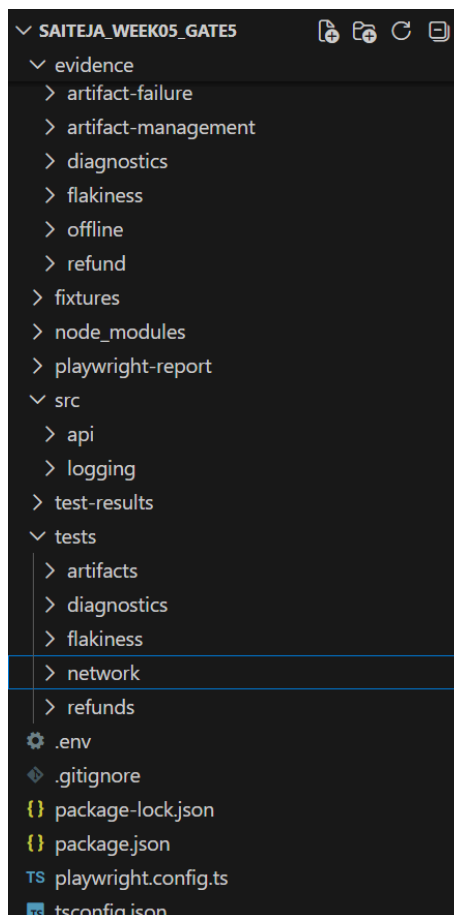
Technology: Playwright with TypeScript

Repository: [ust-sdet-training/SDET at saiteja_week05_gate5](#)

Note:

Meaningful comments have been added throughout the test files to improve readability and make the test flow easier to understand. This report presents each assessment scenario using the **Diagnose → Fix → Evidence → Explain** approach, supported by relevant screenshots.

Code Structure:



1. Objective

The objective of this assessment is to stabilize the Playwright automation suite by identifying failures, diagnosing their root causes, validating fixes, and providing supporting evidence. The assessment focuses on writing reliable tests and demonstrating debugging maturity.

2. Assessment Overview

This assessment covers the following topics:

- Flakiness Analysis
- Diagnostic Logging
- Artifact Management
- Offline & Resilient UI
- Returns & Refund Testing

The overall approach followed throughout the assessment is:

Diagnose → Fix → Evidence → Explain

3. Tools Used

- Playwright
 - TypeScript
 - Node.js
 - Visual Studio Code
 - Git & GitHub
-

4. Flakiness Analysis

Command to run in terminal:

```
npx playwright test makeltFlaky.spec.ts --repeat-each=50
```

Test File: makeltFlaky.spec.ts

Scenario 1 – Race Condition

Issue

The cart total was verified before the asynchronous request completed.

Diagnosis

The assertion executed before the UI finished updating, causing intermittent failures.

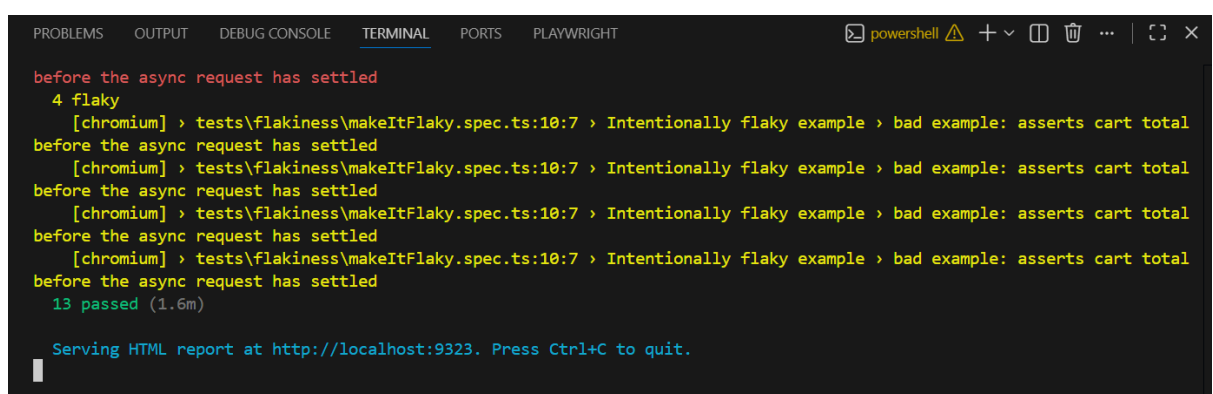
Fix

Use Playwright's web-first assertions and wait for the UI state instead of using very short timeouts.

Evidence

- Screenshot of failed test (Before)

```
15     await page.getByRole("button", { name: "Refresh cart total" }).click();
16
17     // Check the cart total immediately, which is the flaky part.
18     await expect(page.getByTestId("debug-cart-total")).toContainText("Rs. 9,197", {
19         timeout: 620
20     });
21 });
```



- Screenshot of stable execution (After)

```
16
17     // Check the cart total immediately, which is the flaky part.
18     await expect(page.getByTestId("debug-cart-total")).toContainText("Rs. 9,197");
19 });
20
```

```
Running 2 tests using 2 workers
...ss\makeItFlaky.spec.ts:27:7 > Intentionally flaky example > bad example: brittle product locator depends on card position
[INFO] 2026-07-04T11:54:31.733Z - test started
[INFO] 2026-07-04T11:54:31.738Z - Opening catalog page for brittle locator check
...ItFlaky.spec.ts:10:7 > Intentionally flaky example > bad example: asserts cart total before the async request has settled
[INFO] 2026-07-04T11:54:31.895Z - test started
[INFO] 2026-07-04T11:54:31.901Z - Opening debug lab page for flaky cart total check
[INFO] 2026-07-04T11:54:32.643Z - Refreshing cart total
...ss\makeItFlaky.spec.ts:27:7 > Intentionally flaky example > bad example: brittle product locator depends on card position
[INFO] 2026-07-04T11:54:32.643Z - Clicking the travel backpack product link
[INFO] 2026-07-04T11:54:33.423Z - Captured product page heading {"headingText":"Travel Backpack"}
[INFO] 2026-07-04T11:54:33.478Z - test finished {"status":"passed"}
...ItFlaky.spec.ts:10:7 > Intentionally flaky example > bad example: asserts cart total before the async request has settled
[INFO] 2026-07-04T11:54:34.021Z - Captured cart total text {"cartTotalText":"Rs. 9,197"}
[INFO] 2026-07-04T11:54:34.075Z - test finished {"status":"passed"}
2 passed (9.4s)
```

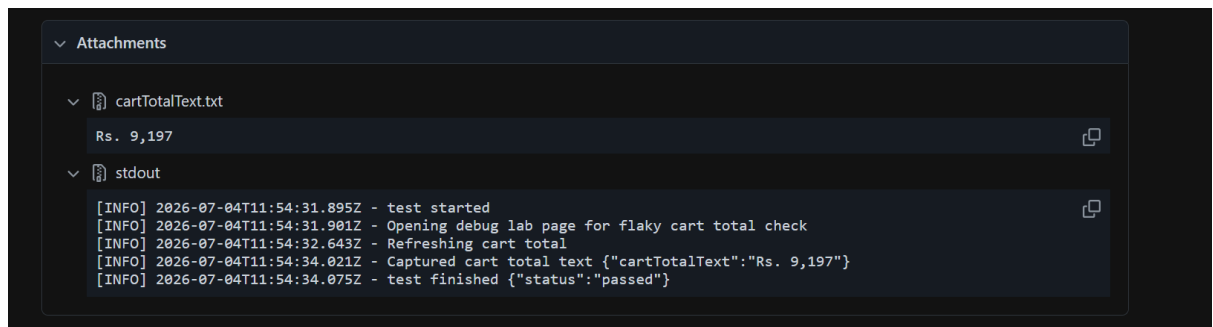
- HTML Report screenshot (Before)

 Intentionally flaky example > bad example: asserts cart total before the async request has settled chromium	13.0s
flakiness/makeltFlaky.spec.ts:10   View Trace	
 Intentionally flaky example > bad example: asserts cart total before the async request has settled chromium	15.7s
flakiness/makeltFlaky.spec.ts:10   View Trace	
 Intentionally flaky example > bad example: asserts cart total before the async request has settled chromium	10.1s
flakiness/makeltFlaky.spec.ts:10   View Trace	
 Intentionally flaky example > bad example: asserts cart total before the async request has settled chromium	9.0s
flakiness/makeltFlaky.spec.ts:10   View Trace	
 Intentionally flaky example > bad example: asserts cart total before the async request has settled chromium	5.6s
flakiness/makeltFlaky.spec.ts:10	
 Intentionally flaky example > bad example: brittle product locator depends on card position chromium	5.8s
flakiness/makeltFlaky.spec.ts:23	
 Intentionally flaky example > bad example: brittle product locator depends on card position chromium	6.1s
flakiness/makeltFlaky.spec.ts:23	
 Intentionally flaky example > bad example: brittle product locator depends on card position chromium	7.6s
flakiness/makeltFlaky.spec.ts:23	
 Intentionally flaky example > bad example: asserts cart total before the async request has settled chromium	6.8s
flakiness/makeltFlaky.spec.ts:10	
 Intentionally flaky example > bad example: brittle product locator depends on card position chromium	3.3s
flakiness/makeltFlaky.spec.ts:23	
 Intentionally flaky example > bad example: brittle product locator depends on card position chromium	4.7s

- HTML Report screenshot (After)

▼ flakiness/makeltFlaky.spec.ts	
 Intentionally flaky example > bad example: asserts cart total before the async request has settled chromium	5.2s
flakiness/makeltFlaky.spec.ts:10	
 Intentionally flaky example > bad example: brittle product locator depends on card position chromium	5.0s
flakiness/makeltFlaky.spec.ts:21	

- Evidence



Scenario 2 – Brittle Locator

Issue

The locator depended on the position of a product card.

Diagnosis

Any UI order change caused the test to fail.

Fix

Use stable locators such as `getByRole()` or `getByTestId()`.

Evidence

- Screenshot of failure (Before)

```
24
25 // Navigate to a product using a fragile position-based selector.
26 await page.locator(".product-grid article:nth-child(2) a").click();
27
28 // Assert that the expected product page is visible.
29 await expect(
30   page.getByRole("heading", { name: "Travel Backpack" })
31 ).toBeVisible();
```

- Screenshot after using stable locator (After)

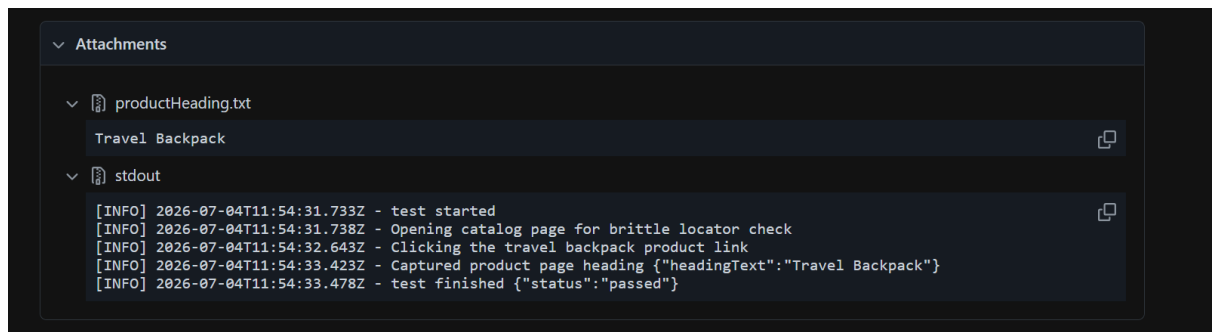
```
24
25 // Navigate to a product using a fragile position-based selector.
26 await page.getByRole("link", { name: "View Travel Backpack" }).click();
27
28 // Assert that the expected product page is visible.
29 await expect(
30   page.getByRole("heading", { name: "Travel Backpack" })
31 ).toBeVisible();
32 });
```

```

Running 2 tests using 2 workers
...ss\makeItFlaky.spec.ts:27:7 > Intentionally flaky example > bad example: brittle product locator depends on card position
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[INFO] 2026-07-04T11:54:33.423Z - Captured product page heading {"headingText":"Travel Backpack"}
[INFO] 2026-07-04T11:54:33.478Z - test finished {"status":"passed"}
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[INFO] 2026-07-04T11:54:34.021Z - Captured cart total text {"cartTotalText":"Rs. 9,197"}
[INFO] 2026-07-04T11:54:34.075Z - test finished {"status":"passed"}
2 passed (9.4s)

```

- Evidence



5. Diagnostic Logging

Command to run in terminal:

npx playwright test diagnostics-logging.spec.ts

Test File: diagnostics-logging.spec.ts

Purpose

To improve debugging by capturing structured logs for every important step in the checkout flow.

Implemented

- Correlation ID
- Structured logging
- Response time logging
- Sensitive data redaction

Benefits

- Easy request tracing

- Faster debugging
- Secure logging without exposing sensitive information

Evidence

- Correlation ID and Redacted log screenshot of the terminal

```
...s\diagnostics\diagnostics-logging.spec.ts:12:7 > Diagnostics and Logging > records a correlated checkout diagnostic trail
[INFO] 2026-07-04T11:27:35.007Z - test started
[INFO] 2026-07-04T11:27:35.020Z - checkout journey started {"cartSession":"w5d2-diagnostics-1783164455019"}
...sts\diagnostics\diagnostics-logging.spec.ts:113:7 > Diagnostics and Logging > redacts sensitive fields in structured logs
[INFO] 2026-07-04T11:27:35.098Z - test started
[INFO] 2026-07-04T11:27:35.103Z - payment payload prepared {"cardNumber":"[REDACTED]","headers":{"authorization":"[REDACTED]"},"orderId":"ORD-LOG-1001","token":"[REDACTED]"}
[INFO] 2026-07-04T11:27:35.123Z - test finished {"status":"passed"}
...s\diagnostics\diagnostics-logging.spec.ts:12:7 > Diagnostics and Logging > records a correlated checkout diagnostic trail
[INFO] 2026-07-04T11:27:35.132Z - cart item added {"cartSession":"w5d2-diagnostics-1783164455019","durationMs":112,"httpStatus":201,"productId":101}
[INFO] 2026-07-04T11:27:35.205Z - order placed {"cartSession":"w5d2-diagnostics-1783164455019","durationMs":24,"httpStatus":201,"orderId":5001,"orderNumber":"ORD-1009","total":4598}
[INFO] 2026-07-04T11:27:35.254Z - test finished {"status":"passed"}
2 passed (7.2s)
```

- HTML Report

Project: chromium4/7/2026, 4:57:30 pm Total time: 7.2s

▼ diagnostics/diagnostics-logging.spec.ts

✓

Diagnostics and Logging › records a correlated checkout diagnostic trail chromium2.3s

diagnostics/diagnostics-logging.spec.ts:12

✓

Diagnostics and Logging › redacts sensitive fields in structured logs chromium2.3s

diagnostics/diagnostics-logging.spec.ts:113

6. Artifact Management

Command to run in terminal:

npx playwright test artifact-management.spec.ts

npx playwright test artifact-failure.spec.ts

Test Files

- artifact-management.spec.ts
- artifact-failure.spec.ts

Purpose

To capture useful debugging evidence whenever tests pass or fail.

Artifacts Captured

- Playwright HTML Report

- Failure Screenshot
- Cart Evidence

Benefits

Developers can understand failures without reproducing the issue.

Evidence

- HTML Report (Success)

captures cart evidence without attaching it on a passing test 2.1s

artifacts/artifact-management.spec.ts:13

chromium

✓ Run

Test Steps

Filter steps

> ✓ Before Hooks	1.2s
> ✓ Navigate to "/debug-lab" — artifacts/artifact-management.spec.ts:17	531ms
> ✓ Expect "toBeVisible" getByRole('heading', { name: 'Debug Lab' }) — artifacts/artifact-management.spec.ts:20	95ms
> ✓ POST "/api/cart/items" — artifacts/artifact-management.spec.ts:23	52ms
> ✓ GET "/api/cart" — artifacts/artifact-management.spec.ts:38	5ms
> ✓ Expect "toBe" — artifacts/artifact-management.spec.ts:47	0ms
> ✓ Expect "toBe" — artifacts/artifact-management.spec.ts:48	1ms
> ✓ Expect "toMatchObject" — artifacts/artifact-management.spec.ts:51	2ms
> ✓ After Hooks	683ms

Attachments

cartResponse.json

```

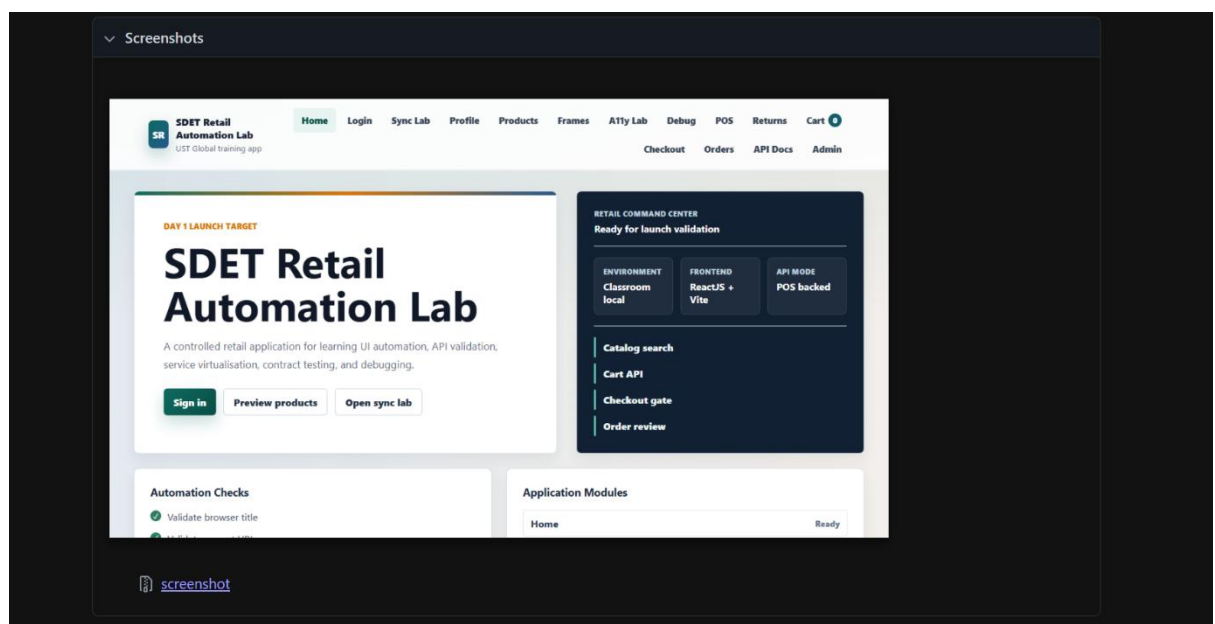
{
  "items": [
    {
      "id": 2,
      "userId": 1,
      "ownerKey": "user-1",
      "productId": 101,
      "quantity": 4,
      "size": "UK 9",
      "color": "Black",
      "fulfilment": "Home delivery",
      "product": {
        "id": 101,
        "name": "Running Shoes",
        "category": "Footwear",
        "price": 4499,
        "stock": 18,
        "brand": "SwiftRun",
        "sku": "FT-SHOE-101",
        "summary": "Lightweight daily trainers with breathable mesh and steady heel support.",
        "tags": [
          "Best seller",
          "Running",
          "COD eligible"
        ]
      }
    }
  ],
  "total": 17996
}

```

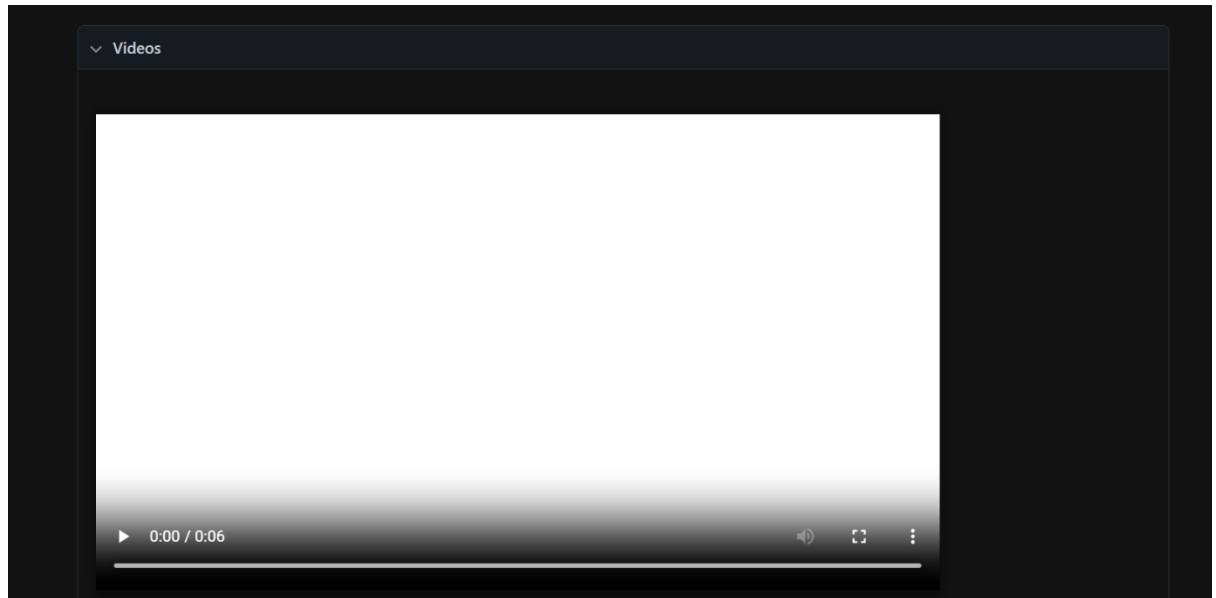

- HTML Report (Failure)



- Failure Screenshot



- Trace Back Screenshot



7. Offline & Resilient UI

Command to run in terminal:

```
npx playwright test network-offline.spec.ts
```

Test File: network-offline.spec.ts

Scenario 1

Verify offline banner.

Scenario 2

Queue sales while offline.

Scenario 3

Synchronize queued sales after reconnecting.

Scenario 4

Rollback optimistic UI after failed synchronization.

Scenario 5

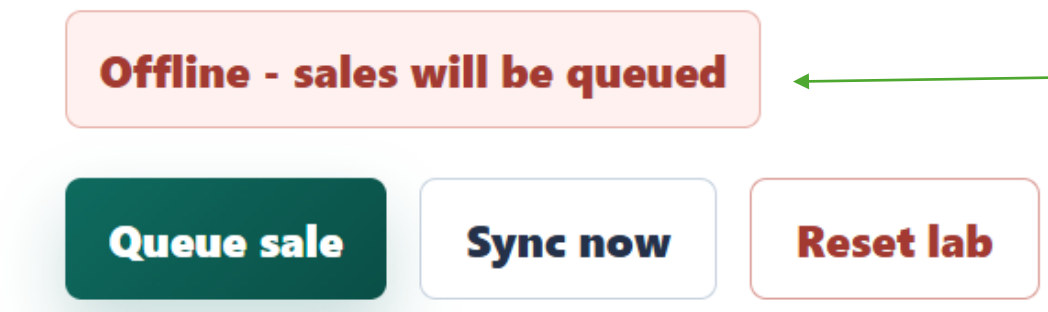
Ensure exactly-once synchronization using Idempotency.

Benefits

The application continues functioning correctly during network interruptions and safely recovers when connectivity is restored.

Evidence

- Offline banner screenshot



- Queue count screenshot



- Sync completed screenshot



Pending outbox: 0



- Rollback screenshot

Sync failed. Optimistic sale rolled back.



Pending outbox: **0**

Last synced sale: SALE-9010

- Exactly-once synchronization screenshot

Local outbox

Queued POS sales



CLIENT SALE ID	PRODUCT	STATUS	TOTAL
sale-1783159737845-a45ga8e4f85	Running Shoes	synced	Rs. 4,499
sale-1783159737057-no4212gvg6	Running Shoes	synced	Rs. 4,499
sale-1783159649279-q24xywyfq59	Running Shoes	synced	Rs. 4,499
sale-1783159648748-r89dj0do947	Running Shoes	synced	Rs. 4,499

8. Returns & Refund Testing

Test File: refund-return.spec.ts

The following business scenarios were validated.

Partial Refund

Command to run in terminal:

`npx playwright test partial-refund.spec.ts`

Verified:

- Refund amount
- Tax calculation
- Ledger update
- UI update

Evidence

- Before refund screenshot of UI

Returnable Lines				
Refund line ledger				
SKU	UNIT	PURCHASED	REFUNDED	RULE
TEE	₹333.00	3	0	RETURNABLE

- After refund screenshot of UI

Returnable Lines				
Refund line ledger				
SKU	UNIT	PURCHASED	REFUNDED	RULE
TEE	₹333.00	3	1	RETURNABLE

- Terminal Output

```
Running 1 test using 1 worker
[chromium] > tests\refunds\partial-refund.spec.ts:13:7 > Partial Refund > should successfully process a partial refund
[INFO] 2026-07-04T10:58:20.250Z - test started
[INFO] 2026-07-04T10:58:20.258Z - Seeding order
[INFO] 2026-07-04T10:58:20.391Z - Opening Returns page {"orderId":7020}
[INFO] 2026-07-04T10:58:21.378Z - Submitting partial refund
[INFO] 2026-07-04T10:58:21.441Z - Ledger verified {"status":"PARTIALLY_REFUNDED","refundCount":1}
[INFO] 2026-07-04T10:58:21.691Z - test finished {"status":"passed"}
1 passed (8.2s)
```

To open last HTML report run:

```
npx playwright show-report
```

```
C:\Users\299172\Documents\Saiteja_week05_gate5>
```

- HTML Report

Partial Refund

should successfully process a partial refund

refunds/partial-refund.spec.ts:13 3.3s

chromium

✓ Run

Test Steps

Filter steps

> ✓ Before Hooks	1.6s
> ✓ POST "/api/refund-lab/orders" — ./src/api/refund.api.ts:14	101ms
> ✓ Expect "toBeTruthy" — ./src/api/refund.api.ts:32	1ms
> ✓ Navigate to "/returns?orderId=7020" — ./src/api/refund.api.ts:107	678ms
> ✓ Expect "toContainText" getByTestId('refund-line') — refunds/partial-refund.spec.ts:32	222ms
> ✓ Expect "toHaveText" getByTestId('refunded-TEE') — refunds/partial-refund.spec.ts:33	65ms
> ✓ POST "/api/refunds" — ./src/api/refund.api.ts:43	10ms
> ✓ Expect "toBeTruthy" — refunds/partial-refund.spec.ts:44	1ms

- Evidence

Attachments

order.json

```
{
  "id": 7020,
  "orderNumber": "RET-7020",
  "ownerKey": "user-1",
  "placedOn": "2026-06-29",
  "daysAgo": 5,
  "returnWindowDays": 30,
  "paymentMethod": "Original card",
  "lines": [
    {
      "sku": "TEE",
      "name": "Training Tee",
      "unitPaise": 33300,
      "qty": 3,
      "refundedQty": 0,
      "finalSale": false,
      "nonReturnable": false
    }
  ],
  "lineTotalPaise": 99900,
  "taxPaise": 4995,
  "totalPaise": 104895,
  "refundableBalancePaise": 104895,
  "refunds": [],
  "refundCount": 0,
  "lastRefund": null,
  "status": "PAID"
}
```

> refund.json

> ledger.json

Full Refund

Command to run in terminal:

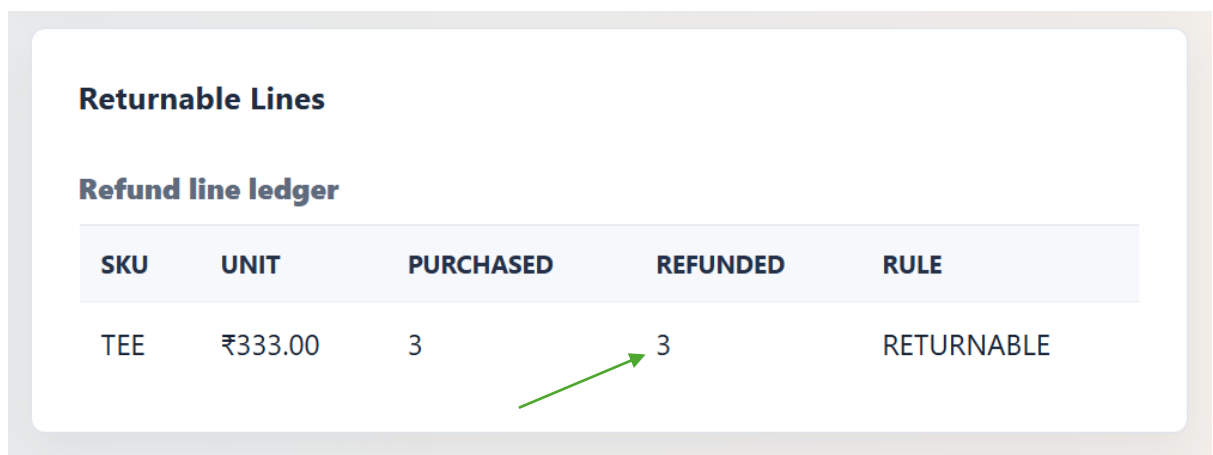
`npx playwright test full-refund.spec.ts`

Verified:

- Complete refund
- Refundable balance becomes zero
- Ledger updated correctly

Evidence

- Full refund screenshot of UI



Returnable Lines				
Refund line ledger				
SKU	UNIT	PURCHASED	REFUNDED	RULE
TEE	₹333.00	3	3	RETURNABLE

- Terminal Output

```
Running 1 test using 1 worker
[chromium] > tests\refunds\full-refund.spec.ts:13:7 > Full Refund > should successfully process a full refund
[INFO] 2026-07-04T11:03:36.847Z - test started
[INFO] 2026-07-04T11:03:36.856Z - Creating order
[INFO] 2026-07-04T11:03:37.020Z - Refunding entire order
[INFO] 2026-07-04T11:03:37.080Z - Full refund completed {"status":"REFUNDED"}
[INFO] 2026-07-04T11:03:38.296Z - test finished {"status":"passed"}
1 passed (7.3s)
```

To open last HTML report run:

```
npx playwright show-report
```

```
C:\Users\299172\Documents\Saiteja_week05_gate5>
```

- HTML Report

Full Refund

should successfully process a full refund

refunds/full-refund.spec.ts:13 3.2s

chromium

✓ Run

Test Steps

Filter steps

> ✓ Before Hooks	1.6s
> ✓ POST "/api/refund-lab/orders" — ./src/api/refund.api.ts:14	120ms
> ✓ Expect "toBeTruthy" — ./src/api/refund.api.ts:32	4ms
> ✓ POST "/api/refunds" — ./src/api/refund.api.ts:43	14ms
> ✓ Expect "toBeTruthy" — refunds/full-refund.spec.ts:36	1ms
> ✓ Expect "toBe" — refunds/full-refund.spec.ts:42	1ms
> ✓ GET "/api/refund-lab/orders/7021" — ./src/api/refund.api.ts:91	8ms
> ✓ Expect "toBeTruthy" — ./src/api/refund.api.ts:98	0ms
> ✓ Expect "toBe" — refunds/full-refund.spec.ts:53	0ms

- Evidence

Attachments

order.json

```
{
  "id": 7021,
  "orderNumber": "RET-7021",
  "ownerKey": "user-1",
  "placedOn": "2026-06-29",
  "daysAgo": 5,
  "returnWindowDays": 30,
  "paymentMethod": "Original card",
  "lines": [
    {
      "sku": "TEE",
      "name": "Training Tee",
      "unitPaise": 33300,
      "qty": 3,
      "refundedQty": 0,
      "finalSale": false,
      "nonReturnable": false
    }
  ],
  "lineTotalPaise": 99900,
  "taxPaise": 4995,
  "totalPaise": 104895,
  "refundableBalancePaise": 104895,
  "refunds": [],
  "refundCount": 0,
  "lastRefund": null,
  "status": "PAID"
}
```

> refund.json

> ledger.json

Refund Eligibility

Command to run in terminal:

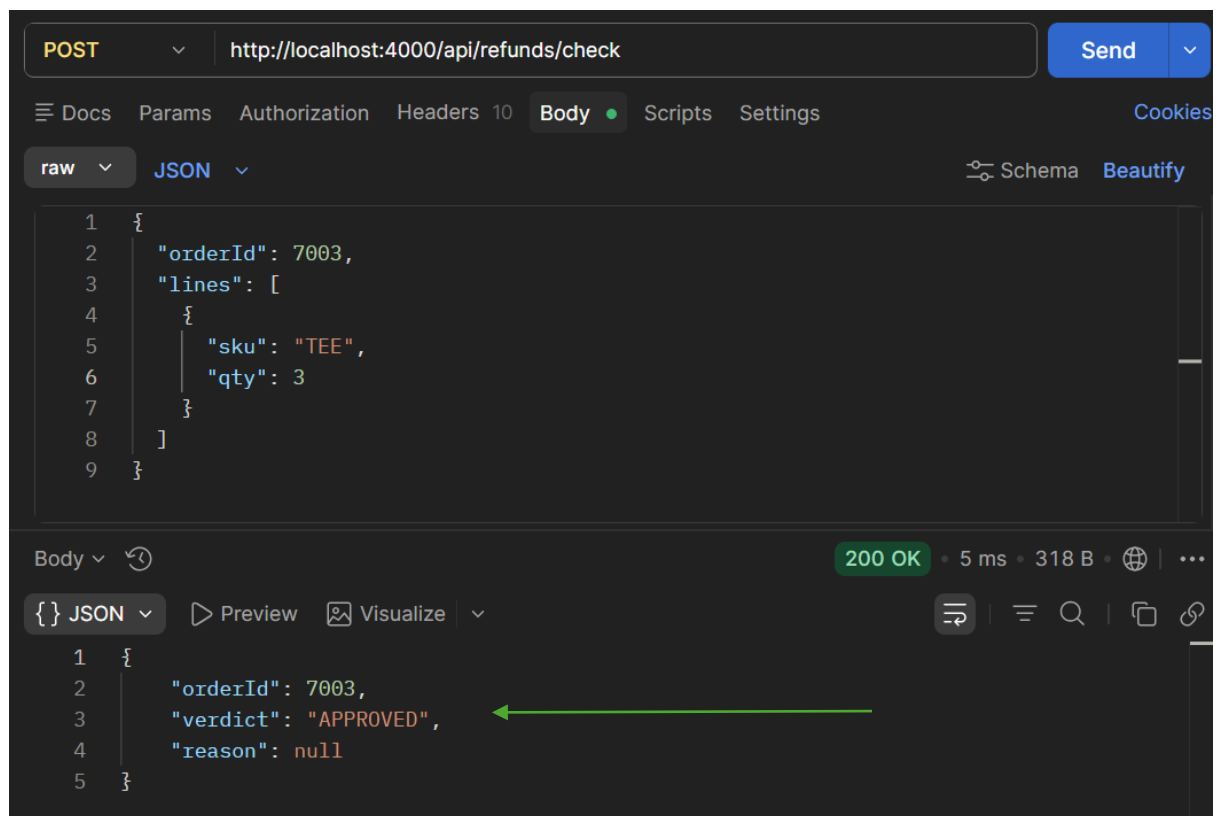
`npx playwright test refund-eligibility.spec.ts`

Verified:

- Valid refund

Evidence

- Refund eligibility screenshot in postman



- Terminal Output

```

[INFO] 2026-07-04T11:05:23.837Z - Refund eligibility verified {"verdict":"OVER_REFUND"}
[INFO] 2026-07-04T11:05:23.870Z - test finished {"status":"passed"}
[chromium] > tests\refunds\refund-eligibility.spec.ts:27:9 > Refund Eligibility > Valid Refund
[INFO] 2026-07-04T11:05:23.961Z - test started
[INFO] 2026-07-04T11:05:23.966Z - Creating order
[INFO] 2026-07-04T11:05:24.056Z - Checking refund eligibility
[INFO] 2026-07-04T11:05:24.089Z - Refund eligibility verified {"verdict":"APPROVED"}
[INFO] 2026-07-04T11:05:24.122Z - test finished {"status":"passed"}
  2 passed (6.5s)

```

To open last HTML report run:

```
npx playwright show-report
```

C:\Users\299172\Documents\Saiteja_week05_gate5>

- HTML Report

Refund Eligibility
next »

Valid Refund

refunds/refund-eligibility.spec.ts:27

2.4s

chromium

✓ Run

Test Steps

Filter steps

Before Hooks
2.2s

POST "/api/refund-lab/orders" — ./src/api/refund.api.ts:14
65ms

Expect "toBeTruthy" — ./src/api/refund.api.ts:32
1ms

POST "/api/refunds/check" — ./src/api/refund.api.ts:66
9ms

Expect "toBeTruthy" — ./src/api/refund.api.ts:82
0ms

Expect "toBe" — refunds/refund-eligibility.spec.ts:52
1ms

After Hooks
936ms

- Evidence

Attachments

order.json

```

{
  "id": 7023,
  "orderNumber": "RET-7023",
  "ownerKey": "user-1",
  "placedOn": "2026-06-29",
  "daysAgo": 5,
  "returnWindowDays": 30,
  "paymentMethod": "Original card",
  "lines": [
    {
      "sku": "TEE",
      "name": "Training Tee",
      "unitPaise": 33300,
      "qty": 3,
      "refundedQty": 0,
      "finalSale": false,
      "nonReturnable": false
    }
  ],
  "lineTotalPaise": 99900,
  "taxPaise": 4995,
  "totalPaise": 104895,
  "refundableBalancePaise": 104895,
  "refunds": [],
  "refundCount": 0,
  "lastRefund": null,
  "status": "PAID"
}

```

refundCheck.json

Rejected Refund

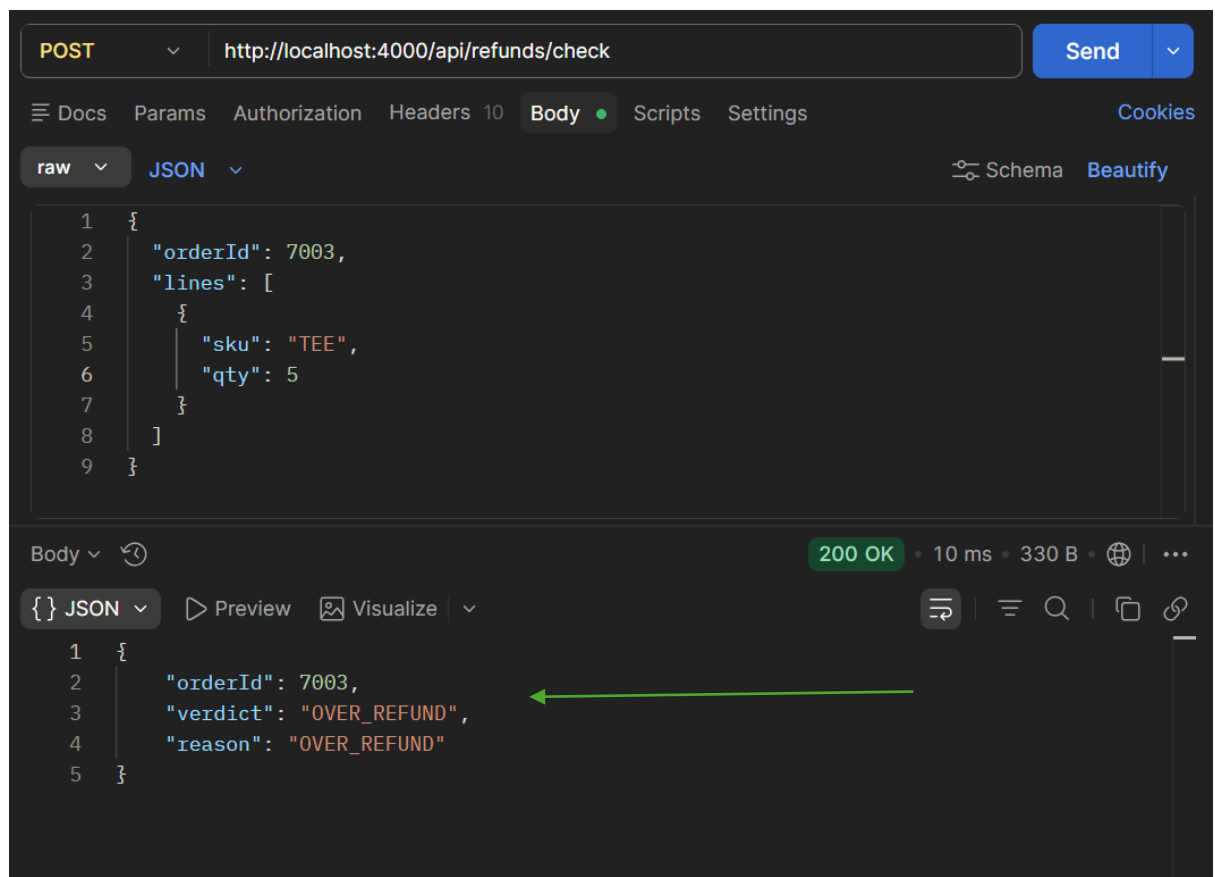
Command to run in terminal:

`npx playwright test rejected-refund.spec.ts`

Verified that over-refund requests are rejected.

Evidence

- Rejected refund screenshot from postman



- Terminal Output

```
Running 1 test using 1 worker
[chromium] > tests\refunds\rejected-refund.spec.ts:11:7 > Rejected Refund > should reject an over-refund request
[INFO] 2026-07-04T11:11:33.337Z - test started
[INFO] 2026-07-04T11:11:33.346Z - Creating order
[INFO] 2026-07-04T11:11:33.520Z - Attempting over refund
[INFO] 2026-07-04T11:11:33.553Z - Over refund rejected {"reason":"OVER_REFUND"}
[INFO] 2026-07-04T11:11:33.591Z - Verified ledger remains unchanged after rejection {"refundableBalancePaise":104895}
[INFO] 2026-07-04T11:11:33.680Z - test finished {"status":"passed"}
1 passed (8.6s)

To open last HTML report run:

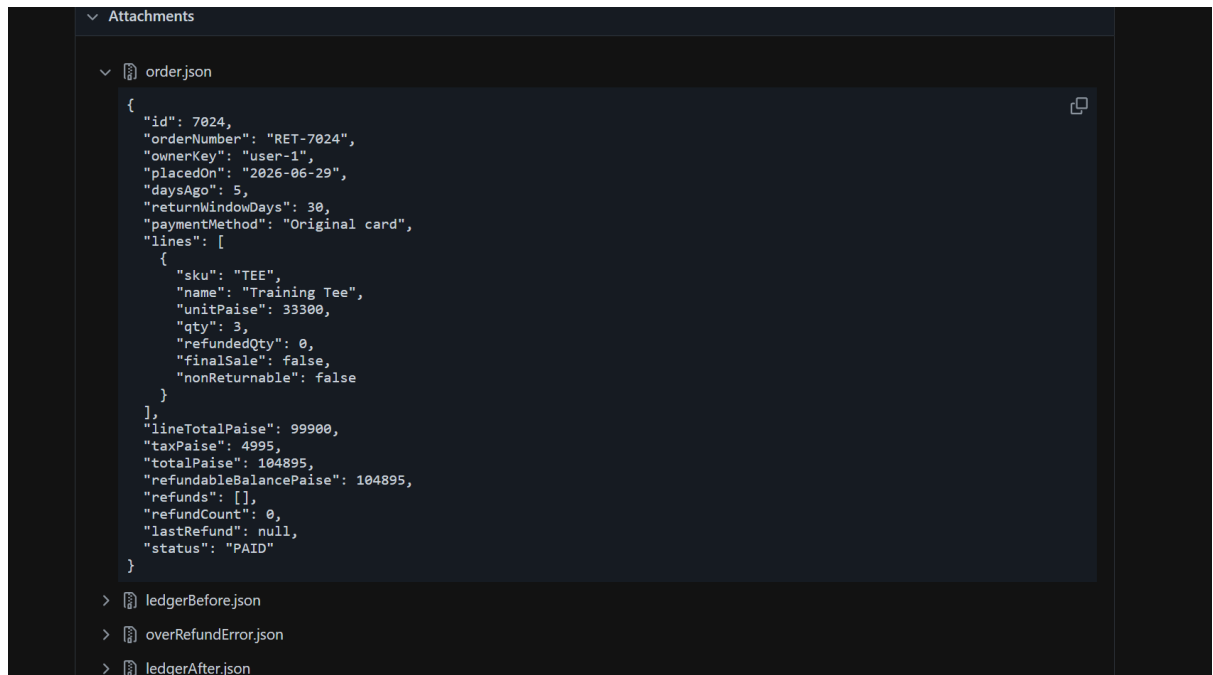
npx playwright show-report
```

- HTML Report

The screenshot shows a Playwright HTML report for a test titled "should reject an over-refund request" located at `refunds/rejected-refund.spec.ts:11`. The test was run using Chromium and passed. The report includes a "Test Steps" section with a search bar and a list of steps, each with a status icon, description, and duration.

Step	Duration
> ✓ Before Hooks	1.8s
> ✓ POST "/api/refund-lab/orders" — ./src/api/refund.api.ts:14	108ms
> ✓ Expect "toBeTruthy" — ./src/api/refund.api.ts:32	2ms
> ✓ GET "/api/refund-lab/orders/7024" — ./src/api/refund.api.ts:91	11ms
> ✓ Expect "toBeTruthy" — ./src/api/refund.api.ts:98	0ms
> ✓ POST "/api/refunds" — ./src/api/refund.api.ts:43	11ms
> ✓ Expect "toBe" — refunds/rejected-refund.spec.ts:39	2ms
> ✓ Expect "toBe" — refunds/rejected-refund.spec.ts:49	1ms
> ✓ Expect "toBe" — refunds/rejected-refund.spec.ts:50	1ms

- Evidence



Duplicate Refund

Command to run in terminal:

npx playwright test duplicate-refund-idempotency.spec.ts

The same refund request was submitted twice.

The application returned the existing refund instead of creating another one.

Evidence

- Duplicate refund
- Screenshot of First time request(Status Code = 201 Created) from Postman

POST http://localhost:4000/api/refunds Send

Docs Params Authorization Headers 11 Body Scripts Settings Cookies

raw JSON Schema Beautify

```
1 {
2   "orderId": 7001,
3   "lines": [
4     {
5       "sku": "TEE",
6       "qty": 2
7     }
8   ]
9 }
```

Body JSON Preview Visualize

201 Created • 10 ms • 638 B •

```
1 {
2   "refundId": 8002,
3   "orderId": 7001,
4   "ownerKey": "user-1",
5   "idempotencyKey": "refund-2026-002",
6   "amountPaise": 69930,
7   "lineAmountPaise": 66600,
8   "taxPaise": 3330,
9   "taxShares": [
10    1665,
11    1665,
12    1665
13  ]
14 }
```

- Terminal Output

```
Running 1 test using 1 worker
...efund-idempotency.spec.ts:12:7 › Duplicate Refund and Idempotency › should process only one refund for duplicate requests
[INFO] 2026-07-04T11:17:13.756Z - test started
[INFO] 2026-07-04T11:17:13.760Z - Creating order
[INFO] 2026-07-04T11:17:13.854Z - Submitting first refund request
[INFO] 2026-07-04T11:17:13.870Z - Submitting duplicate refund request
[INFO] 2026-07-04T11:17:13.901Z - Verified idempotent refund {"refundCount":1,"refundableBalancePaise":69930}
[INFO] 2026-07-04T11:17:13.986Z - test finished {"status":"passed"}
1 passed (5.6s)

To open last HTML report run:

npx playwright show-report
```

- HTML Report

Duplicate Refund and Idempotency

should process only one refund for duplicate requests

refunds/duplicate-refund-idempotency.spec.ts:121.6s

chromium

✓ Run

Test Steps

Q Filter steps

> ✓ Before Hooks1.4s

> ✓ POST "/api/refund-lab/orders" — ./src/api/refund.api.ts:1477ms

> ✓ Expect "toBeTruthy" — ./src/api/refund.api.ts:321ms

> ✓ POST "/api/refunds" — ./src/api/refund.api.ts:438ms

> ✓ Expect "toBeTruthy" — refunds/duplicate-refund-idempotency.spec.ts:380ms

> ✓ POST "/api/refunds" — ./src/api/refund.api.ts:438ms

> ✓ Expect "toBe" — refunds/duplicate-refund-idempotency.spec.ts:531ms

> ✓ Expect "toBe" — refunds/duplicate-refund-idempotency.spec.ts:590ms

> ✓ Expect "toBe" — refunds/duplicate-refund-idempotency.spec.ts:600ms

- Evidence



- Screenshot of Second time request(Status Code = 200 OK) from Postman

POST http://localhost:4000/api/refunds Send

Docs Params Authorization Headers 11 Body Scripts Settings Cookies

raw JSON Schema Beautify

```
1 {
2   "orderId": 7001,
3   "lines": [
4     {
5       "sku": "TEE",
6       "qty": 2
7     }
8   ]
9 }
```

Body ↺

200 OK • 11 ms • 552 B • 🌐 | ⋮

{ } JSON Preview Visualize

```
1 {
2   "refundId": 8002,
3   "orderId": 7001,
4   "ownerKey": "user-1",
5   "idempotencyKey": "refund-2026-002",
6   "amountPaise": 69930,
7   "lineAmountPaise": 66600,
8   "taxPaise": 3330,
9   "taxShares": [
10    1665,
11    1665,
12    1665
13  ]
14 }
```

- Terminal Output

```
Running 1 test using 1 worker
...efund-idempotency.spec.ts:12:7 > Duplicate Refund and Idempotency > should process only one refund for duplicate requests
[INFO] 2026-07-04T11:17:13.756Z - test started
[INFO] 2026-07-04T11:17:13.760Z - Creating order
[INFO] 2026-07-04T11:17:13.854Z - Submitting first refund request
[INFO] 2026-07-04T11:17:13.870Z - Submitting duplicate refund request
[INFO] 2026-07-04T11:17:13.901Z - Verified idempotent refund {"refundCount":1,"refundableBalancePaise":69930}
[INFO] 2026-07-04T11:17:13.986Z - test finished {"status":"passed"}
1 passed (5.6s)

To open last HTML report run:

npx playwright show-report
```

- HTML Report

Duplicate Refund and Idempotency

should process only one refund for duplicate requests

refunds/duplicate-refund-idempotency.spec.ts:12 1.6s

chromium

✓ Run

Test Steps

Filter steps

> ✓ Before Hooks	1.4s
> ✓ POST "/api/refund-lab/orders" — ./src/api/refund.api.ts:14	77ms
> ✓ Expect "toBeTruthy" — ./src/api/refund.api.ts:32	1ms
> ✓ POST "/api/refunds" — ./src/api/refund.api.ts:43	8ms
> ✓ Expect "toBeTruthy" — refunds/duplicate-refund-idempotency.spec.ts:38	0ms
> ✓ POST "/api/refunds" — ./src/api/refund.api.ts:43	8ms
> ✓ Expect "toBe" — refunds/duplicate-refund-idempotency.spec.ts:53	1ms
> ✓ Expect "toBe" — refunds/duplicate-refund-idempotency.spec.ts:59	0ms
> ✓ Expect "toBe" — refunds/duplicate-refund-idempotency.spec.ts:60	0ms

- Evidence

Attachments

order.json

```
{
  "id": 7027,
  "orderNumber": "RET-7027",
  "ownerKey": "user-1",
  "placedOn": "2026-06-29",
  "daysAgo": 5,
  "returnWindowDays": 30,
  "paymentMethod": "Original card",
  "lines": [
    {
      "sku": "TEE",
      "name": "Training Tee",
      "unitPaise": 33300,
      "qty": 3,
      "refundedQty": 0,
      "finalSale": false,
      "nonReturnable": false
    }
  ],
  "lineTotalPaise": 99900,
  "taxPaise": 4995,
  "totalPaise": 104895,
  "refundableBalancePaise": 104895,
  "refunds": [],
  "refundCount": 0,
  "lastRefund": null,
  "status": "PAID"
}
```

> refund.json

> replayRefund.json

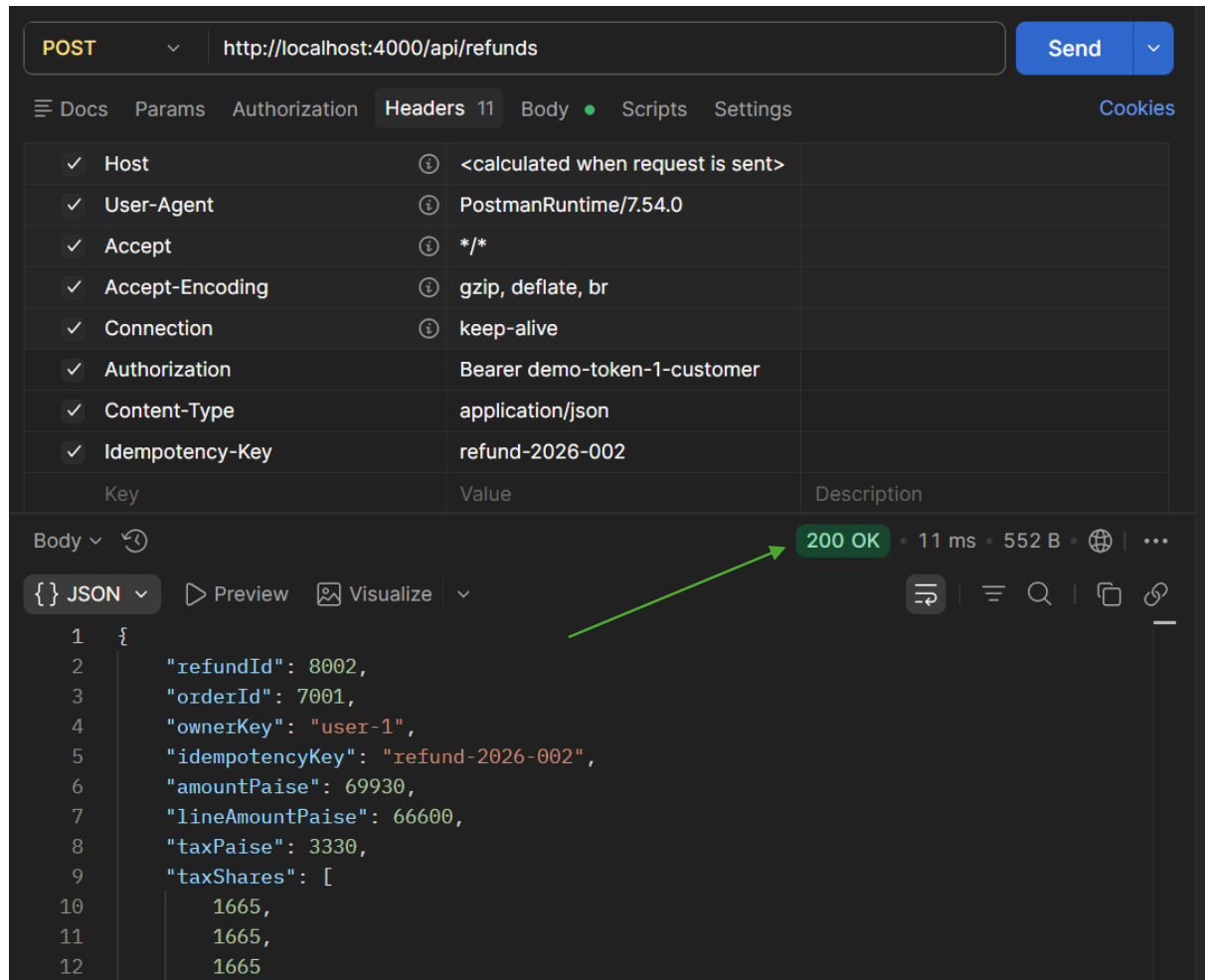
> ledger.json

Idempotency

Verified that duplicate requests with the same idempotency key are processed only once.

Evidence

- Idempotency screenshot



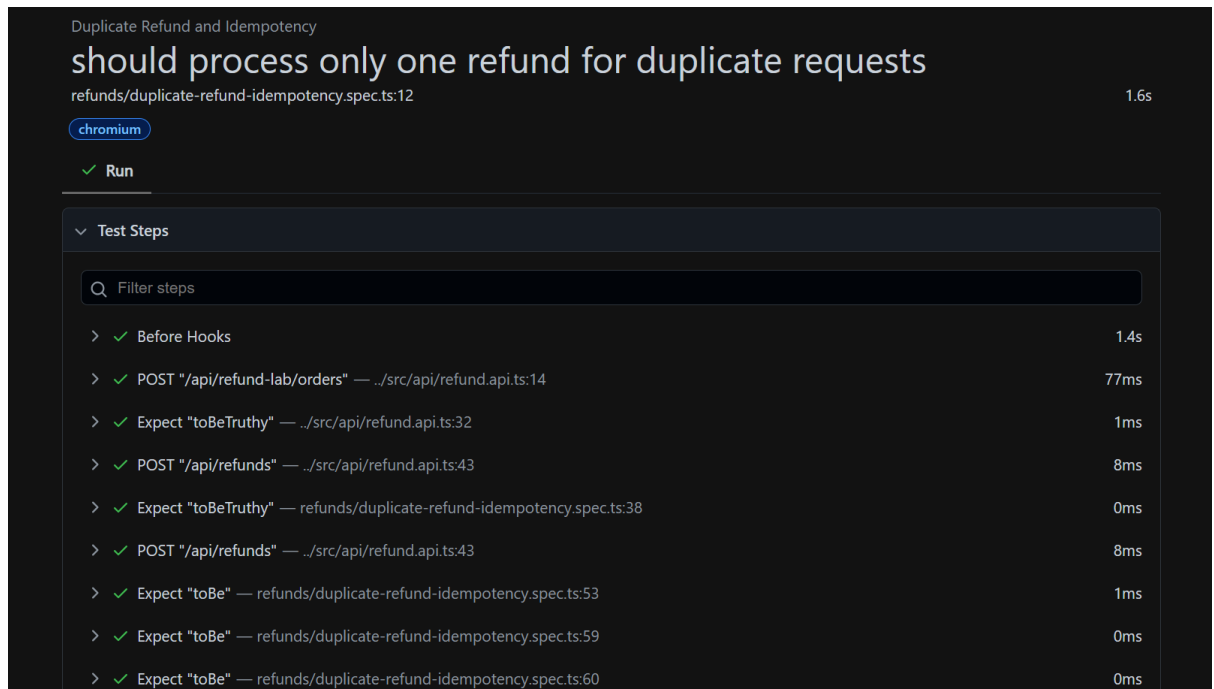
- Terminal Output

```
Running 1 test using 1 worker
...efund-idempotency.spec.ts:12:7 > Duplicate Refund and Idempotency > should process only one refund for duplicate requests
[INFO] 2026-07-04T11:17:13.756Z - test started
[INFO] 2026-07-04T11:17:13.760Z - Creating order
[INFO] 2026-07-04T11:17:13.854Z - Submitting first refund request
[INFO] 2026-07-04T11:17:13.870Z - Submitting duplicate refund request
[INFO] 2026-07-04T11:17:13.901Z - Verified idempotent refund {"refundCount":1,"refundableBalancePaise":69930}
[INFO] 2026-07-04T11:17:13.986Z - test finished {"status":"passed"}
1 passed (5.6s)

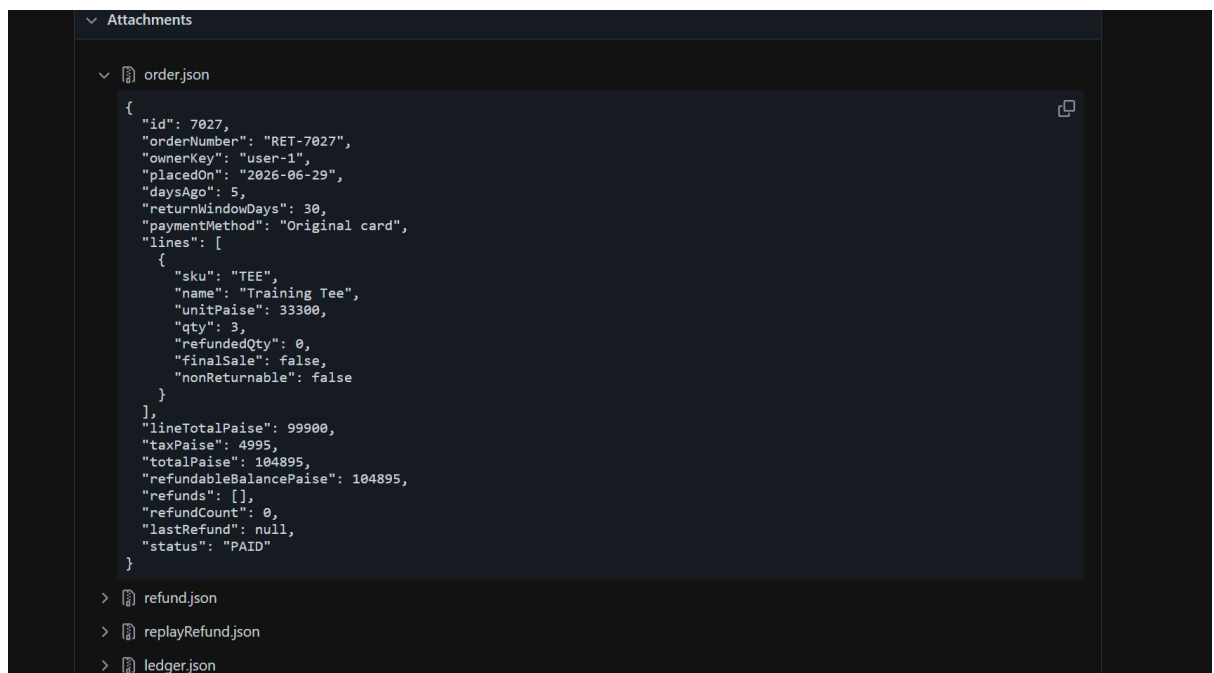
To open last HTML report run:

npx playwright show-report
```

- HTML Report



- Evidence



Money Precision

Verified all monetary calculations using **paise** instead of floating-point values.

Validated:

- Refund Amount

- Tax
- Ledger Balance

Evidence

- Money precision screenshot

```
"amountPaise": 69930,  
"lineAmountPaise": 66600,  
"taxPaise": 3330,  
"taxShares": [  
  1665,  
  1665,  
  1665  
,  
],
```

9. Failure Analysis

Every issue in this assessment follows the same debugging approach.

Symptom

Identify the observed failure.

Diagnosis

Analyze logs, assertions and application behavior to determine the root cause.

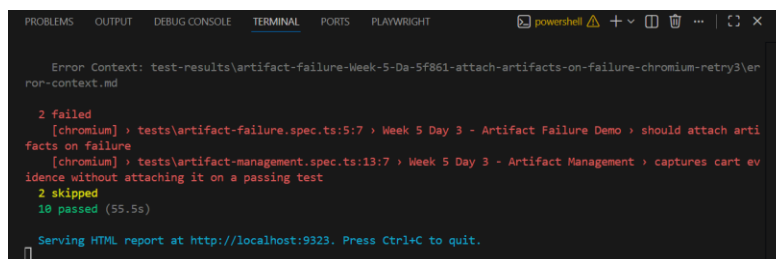
Fix

Apply the appropriate solution instead of temporary workarounds.

Evidence

Validate the fix using:

- Screenshot

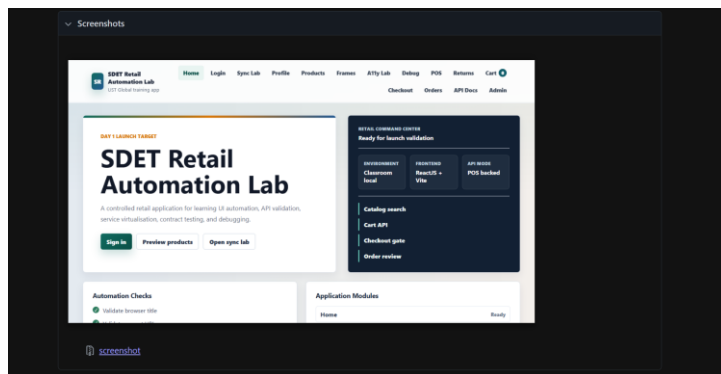


```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS PLAYWRIGHT  
Error Context: test-results\artifact-failure-Week-5-Da-5f861-attach-artifacts-on-failure-chromium-retry3\error-context.md  
2 failed  
[chromium] › tests\artifact-failure.spec.ts:5:7 › Week 5 Day 3 - Artifact Failure Demo › should attach artifacts on failure  
[chromium] › tests\artifact-management.spec.ts:13:7 › Week 5 Day 3 - Artifact Management › captures cart evidence without attaching it on a passing test  
2 skipped  
10 passed (55.5s)  
Serving HTML report at http://localhost:9323. Press Ctrl+C to quit.
```

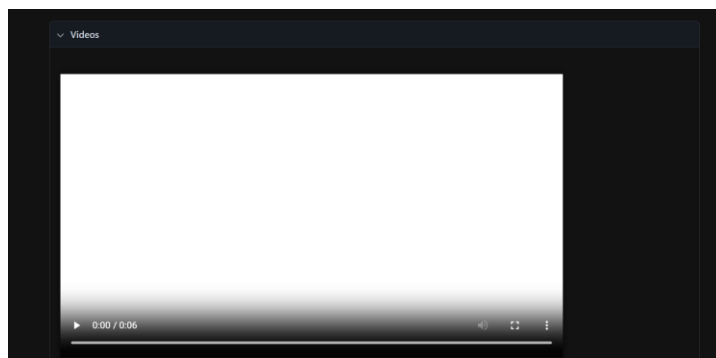
- HTML Report



- Failure Screenshot



- Trace Back Video



10. Debugging Maturity

This assessment demonstrates debugging maturity through:

- Identifying flaky tests
- Diagnosing root causes
- Structured diagnostic logging
- Artifact collection
- Offline resilience testing

- Business rule validation
 - Refund validation
 - Idempotency testing
 - Financial validation using paise
-

11. Challenges Faced

- Diagnosing flaky tests
 - Handling offline scenarios
 - Validating idempotent requests
 - Preventing duplicate refunds
 - Testing financial calculations
 - Capturing useful debugging evidence
-

12. Conclusion

This assessment successfully demonstrates stable Playwright automation by focusing on diagnosing issues, implementing reliable solutions, collecting meaningful evidence, and clearly explaining the results. The project validates critical business scenarios such as offline operations, refund processing, artifact management, and structured diagnostic logging while improving overall debugging maturity.

13. Evidence Summary

Assessment Area	Evidence to Include
Flakiness Analysis	Before failure screenshot, After fix screenshot, HTML Report
Diagnostic Logging	Correlation ID logs, Structured logs, Redacted logs, Trace flow screenshot
Artifact Management	HTML Report (Pass), HTML Report (Fail), Failure Screenshot, Trace Back Screenshot

Assessment Area	Evidence to Include
Offline & Resilient UI	Offline banner, Queue count, Rollback, Sync success
Returns & Refund Testing	Partial refund, Full refund, Rejected refund, Duplicate refund, Idempotency, Money precision